

* Phase Shift Keying (PSK)

- It is a digital modulation technique in which the phase of a carrier signal is varied according to the digital data, while the amplitude and frequency remain constant.

* Binary PSK:

→ In Binary PSK:

- Binary 1 → Carrier phase = 0°
- Binary 0 → Carrier phase = 180°

* Carrier signal:

- A high frequency sinusoidal signal used to transmit information.

* Modulating signal:

- A digital pulse signal that controls the phase change of carrier

* Phase reversal

- A sudden change of carrier phase by 180°

* Advantages

- High noise immunity
- Constant amplitude reduces distortion
- Efficient use of bandwidth
- Better than ASK and PDK.

* Disadvantages:

- Circuit complexity is high

- More complex demodulation
- Cost is higher compared to ASK.

* Applications

- Satellite communication
- Wi-fi systems
- Bluetooth
- Cellular communication

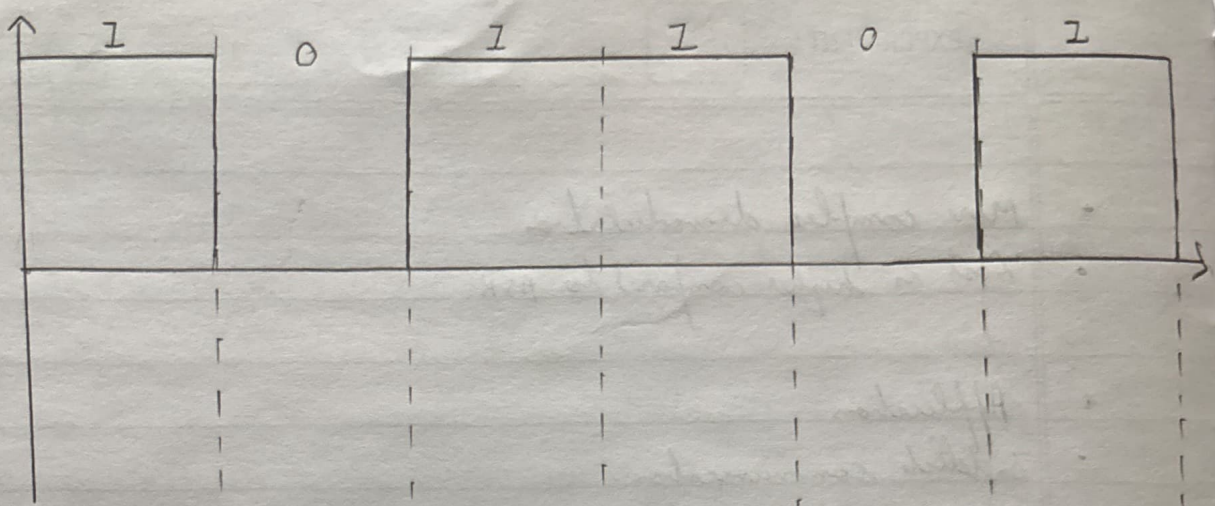
* Components Required

- 741 operational Amplifier
- LF398N IC
- Resistors
- Sine waves
- Digital pulse generator
- Power supply.
- Connecting wires

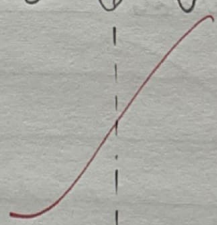
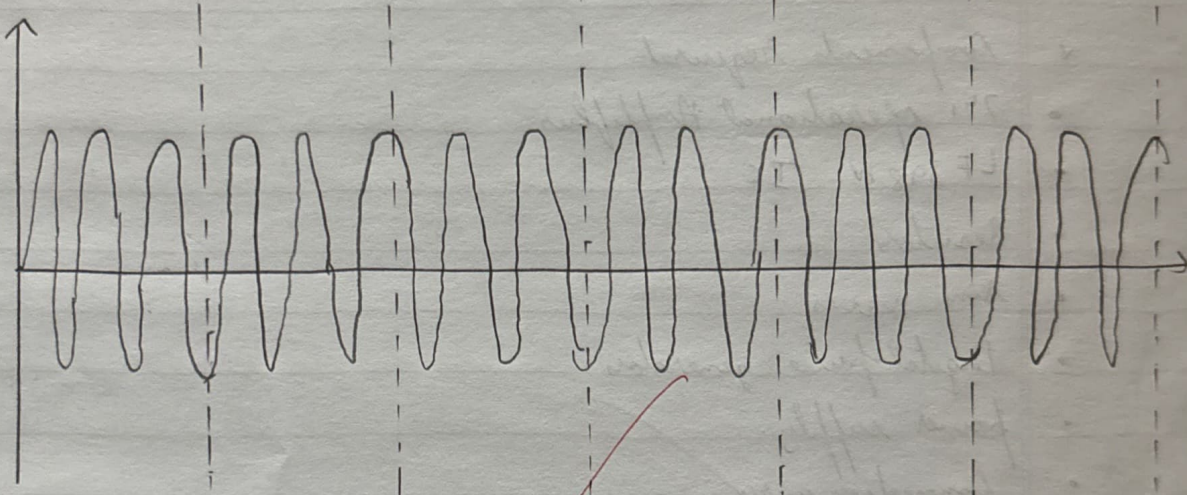
* Procedure

- Apply the carrier sine waves to the ~~so~~ input of the 7410A.
- Use the op. amp to generate two phase-shifted signals
- Apply the digital message signal to control the switching circuit
- Power the IC's using the specified supply voltages.
- Connect a CRO probe to observe.
- Switch ON the power supply and observe waves
- Switch OFF the power supply after observation

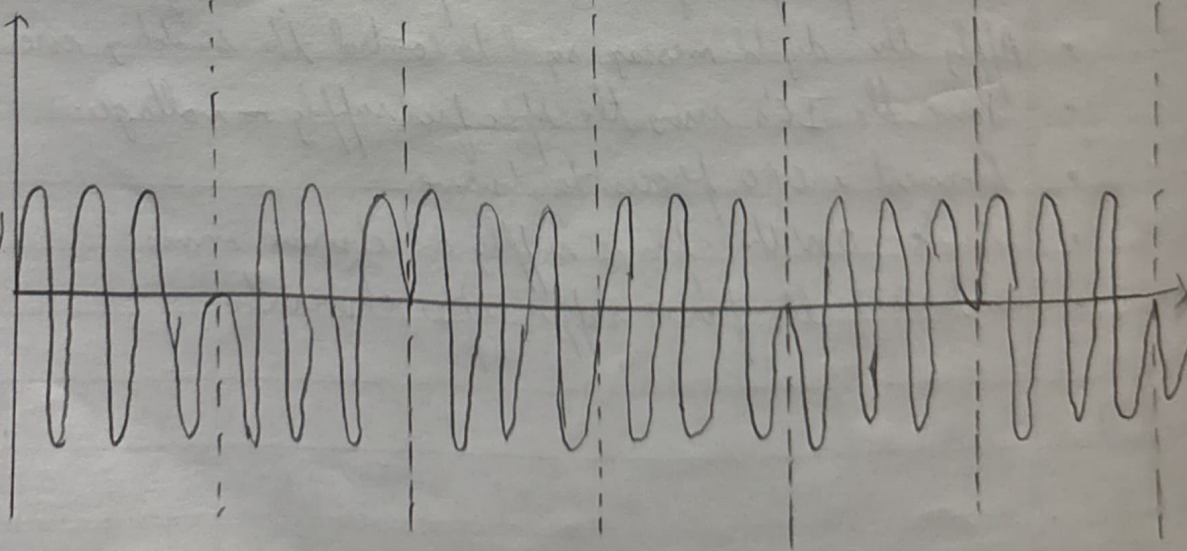
Digital
Signal



Carrier
Signal



PSK
Modulated
Signal



Observation:

when message signal is logic 1, the carrier phase remains unchanged (0°).

when message signal is logic 0, the carrier phase shifts by 180° .

The amplitude and frequency of the carrier remains constant

Result:

PSK was successfully implemented and the PSK waveform showing phase reversals according to the digital input was observed on the CRO.

Conclusion:

The experiment the working of a Wi-Fi network, where multiple wireless devices communicate through an access point

Packet capture shows successful data transmission, proving that wifi provides reliable and flexible wireless for LANs.

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